

1-1 *Relative asymptotic growths*

Indicate, for each pair of expressions (A, B) in the table below, whether A is O , o , Ω , ω , or Θ of B . Assume that $k \geq 1$, $\epsilon > 0$, and $c > 1$ are constants. Your answer should be in the form of the table with “yes” or “no” written in each box.

	A	B	O	o	Ω	ω	Θ
<i>a.</i>	$\lg^k n$	n^ϵ					
<i>b.</i>	n^k	c^n					
<i>c.</i>	$\sqrt{n}$	$n^{\sin n}$					
<i>d.</i>	2^n	$2^{n/2}$					
<i>e.</i>	$n^{\lg c}$	$c^{\lg n}$					
<i>f.</i>	$\lg(n!)$	$\lg(n^n)$					

1-2 *Ordering by asymptotic growth rates*

- a.* Rank the following functions by order of growth; that is, find an arrangement $g_1, g_2, \dots, g_{30}$ of the functions satisfying $g_1 = \Omega(g_2)$, $g_2 = \Omega(g_3)$, $\dots$, $g_{29} = \Omega(g_{30})$. Partition your list into equivalence classes such that functions $f(n)$ and $g(n)$ are in the same class if and only if $f(n) = \Theta(g(n))$.

$\lg(\lg^* n)$	$2^{\lg^* n}$	$(\sqrt{2})^{\lg n}$	n^2	$n!$	$(\lg n)!$
$(\frac{3}{2})^n$	n^3	$\lg^2 n$	$\lg(n!)$	2^{2^n}	$n^{1/\lg n}$
$\ln \ln n$	$\lg^* n$	$n \cdot 2^n$	$n^{\lg \lg n}$	$\ln n$	1
$2^{\lg n}$	$(\lg n)^{\lg n}$	e^n	$4^{\lg n}$	$(n+1)!$	$\sqrt{\lg n}$
$\lg^*(\lg n)$	$2^{\sqrt{2 \lg n}}$	n	2^n	$n \lg n$	$2^{2^{n+1}}$

1-3 *Asymptotic notation properties*

Let $f(n)$ and $g(n)$ be asymptotically positive functions. Prove or disprove each of the following conjectures.

- a.* $f(n) = O(g(n))$ implies $g(n) = O(f(n))$.
- b.* $f(n) + g(n) = \Theta(\min(f(n), g(n)))$.
- c.* $f(n) = O(g(n))$ implies $\lg(f(n)) = O(\lg(g(n)))$, where $\lg(g(n)) \geq 1$ and $f(n) \geq 1$ for all sufficiently large n .
- d.* $f(n) = O(g(n))$ implies $2^{f(n)} = O(2^{g(n)})$.

1-4 Consider sorting n numbers stored in array A by first finding the largest element of A and exchanging it with the element in $A[n]$. Then find the second largest element of A , and exchange it with $A[n-1]$. Continue in this manner for all n elements of A . Write pseudocode for this algorithm, and answer the following questions: What loop invariant does this algorithm maintain? Give the best-case and worst-case running times of selection sort in asymptotic notation.